



Optimization of the one stage cycloidal gearbox as a non-linear least squares problem

Roman Król¹[0000-0002-6279-9562], Marcin Wikło²[0000-0001-5210-6657],
Krzysztof Olejarczyk³[0000-0002-9055-161X], Krzysztof Kołodziejczyk⁴[0000-0002-2105-293X],
Albert Zieja⁵[0000-0002-1990-4531]

University of Technology and Humanities in Radom, Faculty of Mechanical Engineering
ul. Stasieckiego 54, 26-600 Radom, Poland

¹r.krol@uthrad.pl, ²m.wiklo@uthrad.pl, ³k.olejarczyk@uthrad.pl,
⁴k.kolodziejczyk@uthrad.pl, ⁵a.zieja@uthrad.pl

Abstract. Cycloidal gearboxes are mechanisms which can work under higher loading conditions than other sprockets with comparable dimensions. Cycloidal gears can be smaller than other gears because in these complex mechanisms many lobes work at the same time. In this article, Levenberg-Marquardt algorithm for solving non-linear least squares problems is used for optimization of the one stage cycloidal gearbox. The results are compared with the Steepest Descent Method applied for the same theoretical model of the optimized gear. Application of gradient based optimization method for cycloidal gears is innovative. In other works, a genetical algorithm approach is used, but there are many simplifications in an objective function and constraints concerning the shape of the cycloidal gear. In this paper, the function for the accurate calculation of the cycloidal gear volume is presented. Optimization method was implemented in MATLAB software. In the presented method: the volume of the cycloidal gear, the contact stress in the internal sleeves, the contact stress at the lobes and the contact stress at the pits of the cycloidal gear are minimized with the box constraints implemented using the Penalty Function Method.

Keywords: Cycloidal gearbox, Levenberg-Marquardt algorithm, Steepest Descent Method, Optimization, Penalty Function Method

1 Introduction

The cycloidal gearboxes have been intensively developing during the last years. Their history was connected at the beginning with machines where large speed reduction ratio was desired. The advantages of the gearboxes connected with large reduction ratio achievable in one stage, small size, possibility to an overload of the gear train without destroying of the mechanism and what is more important possibility to have almost zero backlashes [1], [2] introduced the cycloidal gearboxes to robotics market. The concept of the cycloidal gear design is investigated in many papers including efficiency, operating conditions, cycloidal disc profile with no-undercutting condition as well as with new design concepts [3]–[5].

In the robotics applications where the small size of the gearbox is needed there are used harmonic gears and generally, there are no other alternatives. This is the main reason to undertake the problem of the optimization of the cycloidal gear. The discussion regarding the unified approach to optimization in point of view of basic aspects of the gear design where presented in [6]. The optimal cycloidal gear design where undertaken in [7], [8] where also multi-objective optimization where introduced.

2 Theoretical model assumptions

In this paper the theoretical model of a cycloidal gearbox is optimized using the Levenberg-Marquardt algorithm and the Steepest Descent Method. Theoretical model consists of the single cycloidal wheel with 15 lobes, 16 external sleeves and 8 internal sleeves. The MATLAB functions can be easily changed to optimize gears with another number of lobes, external and internal sleeves [9]. The function used for volume calculation takes into account only mentioned parts without input and output shafts. The theoretical model is shown in Fig. 1.

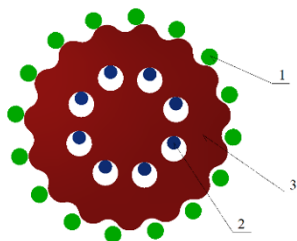


Fig. 1. The theoretical model of the gearbox: the external sleeve (1), the internal sleeve (2) and the cycloidal wheel (3)

The Levenberg-Marquardt algorithm is used for a non-linear least squares problems. This algorithm has been chosen because it is faster than the Steepest Descent Method. The purpose of this algorithm is to find the local minimum of the objective function. The residual vector used in the Levenberg-Marquardt algorithm is shown in equation (1).

$$r(x) = \begin{bmatrix} V(x) & \sigma_{EXLOBE}(x) & \sigma_{EXPIT}(x) & \sigma_{IN}(x) & P_{V1}(x) & P_{E1}(x) & \dots \\ P_{M1}(x) & P_{M2}(x) & P_{H1}(x) & P_{RS1}(x) & P_{RH1}(x) & P_{Q1}(x) & \dots \\ P_{RW1}(x) & P_{RW2}(x) & P_{\rho P1}(x) & P_{\rho P2}(x) & P_{\rho N1}(x) & P_{\rho N2}(x) & \dots \\ P_{SIGEXL}(x) & P_{SIGEXP}(x) & P_{SIGIN}(x) \end{bmatrix}^T \quad (1)$$

where: x – the solution vector, $V(x)$ – the volume of the gearbox (cycloidal wheel, internal and external sleeves), $\sigma_{EXLOBE}(x)$ – the contact stress between the external

sleeve and the lobe, $\sigma_{\text{EXPIT}}(x)$ – the contact stress between the external sleeve and the pit (Fig. 2), $\sigma_{\text{IN}}(x)$ – the contact stress between the internal sleeve and cycloidal wheel, and the penalty functions "P", where the "1" and "2" indexes are for the lower and upper boundaries respectively: $P_{V1}(x)$ – for the cycloidal gearbox volume, $P_{E1}(x)$ – for the eccentricity factor, $P_{M1}(x)$, $P_{M2}(x)$ – for the short-width coefficient, $P_{H1}(x)$ – for the gear thickness (all parts have the same thickness), $P_{RS1}(x)$ – for the internal sleeve radius, $P_{RH1}(x)$ – for the hole corresponding to the internal sleeve (Fig. 1), $P_{Q1}(x)$ – for the external sleeve radius, $P_{RW1}(x)$, $P_{RW2}(x)$ – for the cycloidal gear holes location radius, $P_{pP1}(x)$, $P_{pP2}(x)$ – for the lobe curvature radius, $P_{pN1}(x)$, $P_{pN2}(x)$ – for the pit curvature radius, $P_{\text{SIGEXL}}(x)$ – upper limit of the contact stress between the lobe and the external sleeve, $P_{\text{SIGEXP}}(x)$ – upper limit of the contact stress between the pit and the external sleeve, $P_{\text{SIGIN}}(x)$ – upper limit of the contact stress between an internal sleeve and the cycloidal wheel.

The solution vector shown in equation (2) contains parameters of the cycloidal gear parametric equations (3) and (4). It also contains additional parameters concerning radiuses of the sleeves, holes and the holes location. Number of lobes (z_s) and number of external (z_k) and internal (z_i) sleeves are constant for the given optimization issue. There are also constant material parameters: Poisson's ratios and Young Modules for sleeves and cycloidal gear. The optimization framework can be successfully used for a multi-material analysis, where sleeves are made of one material and a cycloidal gear is made of another.

$$x=[e, q, m, h, R_s, R_h, R_w], \quad (2)$$

$$u(\alpha) = \frac{e \cdot (z_s + 1)}{m} \cos(\alpha) + e \cdot \cos(z_k \cdot \alpha) - q \cdot \cos\left(\alpha + \text{atan}\left(\frac{\sin(z_s \cdot \alpha)}{\frac{1}{m} + \cos(z_s \cdot \alpha)}\right)\right) \quad (3)$$

$$v(\alpha) = \frac{e \cdot (z_s + 1)}{m} \sin(\alpha) + e \cdot \sin(z_k \cdot \alpha) - q \cdot \sin\left(\alpha + \text{atan}\left(\frac{\sin(z_s \cdot \alpha)}{\frac{1}{m} + \cos(z_s \cdot \alpha)}\right)\right) \quad (4)$$

where: e – eccentricity ratio, q – external sleeve radius, m – short-width coefficient, h – cycloidal gear thickness, R_s – internal sleeve radius, R_h – radius of the hole corresponding to the internal sleeve, R_w – radius of the holes location, $u(\alpha)$ – horizontal coordinate in the cartesian system, in which the parametric equations are defined, $v(\alpha)$ – vertical coordinate in the cartesian system, in which the parametric equations are defined.

3 Function for the numerical calculation of the cycloidal gearbox volume

In this paper volume of the cycloidal gearbox is minimized. To perform the optimization, the objective function should be constrained. The contact stresses between sleeves and cycloidal gear increase with the decrease of the gear volume. If it is going to be used the gradient optimization method, than this fact imposes the use of exact function defining the volume of the cycloidal gear from the parametric equations. Jacobi matrix

should contain the derivatives of the parametric equations, which can be determined by the numeric differentiation of the exact volume function. In the case when a cycloidal gear is modeled as a circle with pitch diameter, the same values of volume could be obtained for the various curvature radiuses of the pits and lobes.

The volume of a cycloidal wheel on the basis of its parametric equations can be solved from (5) and the volume of a cycloidal gearbox from (6).

$$V_{WHEEL} = h \cdot \int_0^{2\pi} |v(\alpha)| \cdot \left| \frac{du(\alpha)}{d\alpha} \right| d\alpha \quad (5)$$

$$V_{GEAR} = V_{WHEEL} + z_k \cdot V_{EX_SLEEVE} + z_i \cdot V_{IN_SLEEVE} - z_i \cdot V_{IN_HOLE} \quad (6)$$

where: V_{WHEEL} – the volume of the cycloidal wheel, h – the thickness of the gear, V_{EX_SLEEVE} – the volume of the single external sleeve, V_{IN_SLEEVE} – the volume of the single internal sleeve, V_{IN_HOLE} – the volume of the single hole in the cycloidal wheel. The derivative of the parametric equation $u(\alpha)$ is calculated using symbolic computation in the MATLAB software. It is given in equation (7).

$$\begin{aligned} S &= \sin(\alpha \cdot z_s), C = \cos(\alpha \cdot z_s) + \frac{1}{m} \\ A &= e \cdot z_k \cdot \sin(\alpha \cdot z_k), B = \frac{e \cdot \sin(\alpha) \cdot (z_s + 1)}{m} \\ \frac{du(\alpha)}{d\alpha} &= q \cdot \sin\left(\alpha + \operatorname{atan}\left(\frac{S}{C}\right)\right) \cdot \left(\frac{\frac{z_s \cdot S^2}{C^2} + \frac{z_s \cdot \cos(\alpha \cdot z_s)}{C}}{\frac{S^2}{C^2} + 1} + 1\right) - A - \\ &B \sin\left(\alpha + \operatorname{atan}\left(\frac{\sin(\alpha \cdot z_s)}{\cos(\alpha \cdot z_s) + \frac{1}{m}}\right)\right) \cdot \left(\frac{\frac{z_s \cdot \sin(\alpha \cdot z_s)^2}{(\cos(\alpha \cdot z_s) + \frac{1}{m})^2} + \frac{z_s \cdot \cos(\alpha \cdot z_s)}{\cos(\alpha \cdot z_s) + \frac{1}{m}}}{\frac{\sin(\alpha \cdot z_s)^2}{(\cos(\alpha \cdot z_s) + \frac{1}{m})^2} + 1} + 1\right) - \\ &\frac{14 \cdot z_k \cdot \sin(\alpha \cdot z_k)}{5} - \frac{e \cdot \sin(\alpha) \cdot (z_s + 1)}{m} \quad (7) \end{aligned}$$

Volume of a cycloidal wheel is calculated numerically using the instructions (8)-(12) shown below:

$$dt = \frac{2\pi}{N} \quad (8)$$

$$V_{GEAR} = 0 \quad (9)$$

$$\alpha = i \cdot dt + \frac{dt}{2} \quad (10)$$

$$A_{WHEEL} = \sum_{i=0}^{N-1} |v(\alpha)| \cdot \left| \frac{du(\alpha)}{d\alpha} \right| \quad (11)$$

$$V_{WHEEL} = A_{WHEEL} \cdot h \quad (12)$$

where: N – number of integration points, dt – the numerical interval, α - independent variable of the integrand. A_{WHEEL} – area of the cycloidal wheel.

4 The optimization algorithm

Two algorithms are compared in the optimization of the cycloidal gearbox: the Levenberg-Marquardt algorithm and the Steepest Descent Method. The first algorithm is faster than the second, but the Steepest Descent Method guarantee the convergence. The accuracy in that method depends on the step interval and constants which scale the algorithm behavior. The Levenberg-Marquardt algorithm instructions (13)-(15) are shown below:

$$x_{i+1} = x_i - (H(x_i) + \lambda \cdot \text{diag}[H(x_i)])^{-1} \nabla r(x_i) \quad (13)$$

$$\text{diag}[H(x_i)] = \begin{bmatrix} h_{11} & & 0 & & 0 \\ & \ddots & & & \\ 0 & & h_{ii} & & 0 \\ & & & \ddots & \\ 0 & & 0 & & h_{nn} \end{bmatrix} \quad (14)$$

$$H(x_i) = J^T(x_i) \cdot J(x_i) \quad (15)$$

where: x – the solution vector, $H(x)$ – Hessian matrix (the second derivatives matrix), $r(x)$ – the residual vector, $J(x)$ – Jacobi matrix (the first derivatives matrix), λ – the coefficient, which decides about the step size.

Following instruction (16) present the Steepest Descent Method, where α – the coefficient, which decides about the step size.

$$x_{i+1} = x_i - \alpha \cdot \nabla r(x_i) \quad (16)$$

4.1 The objective function, its Jacobi matrix and the constraints

The objective function is given in section 1 as the residual vector $r(x)$. The Levenberg-Marquardt algorithm requires the definition of the function which determines Jacobi matrix of the objective function. That function is also used to determine gradient in the Steepest Descent Method which is shown in equation (17):

$$\nabla r(x) = J(x)^T \cdot r(x), \quad (17)$$

The $J(x)$ function contains the derivatives of the volume, the stress and the penalty functions which are placed in the residual vector. The $J(x)$ function is given in equations (18) and (19).

$$J(x) = \begin{bmatrix} \frac{\partial r_1(x)}{\partial e} & \frac{\partial r_1(x)}{\partial q} & \frac{\partial r_1(x)}{\partial m} & \frac{\partial r_1(x)}{\partial h} & \frac{\partial r_1(x)}{\partial R_s} & \frac{\partial r_1(x)}{\partial R_h} & \frac{\partial r_1(x)}{\partial R_w} \\ \vdots & & & & & & \\ \frac{\partial r_i(x)}{\partial e} & \frac{\partial r_i(x)}{\partial q} & \frac{\partial r_i(x)}{\partial m} & \frac{\partial r_i(x)}{\partial h} & \frac{\partial r_i(x)}{\partial R_s} & \frac{\partial r_i(x)}{\partial R_h} & \frac{\partial r_i(x)}{\partial R_w} \\ \vdots & & & & & & \\ \frac{\partial r_n(x)}{\partial e} & \frac{\partial r_n(x)}{\partial q} & \frac{\partial r_n(x)}{\partial m} & \frac{\partial r_n(x)}{\partial h} & \frac{\partial r_n(x)}{\partial R_s} & \frac{\partial r_n(x)}{\partial R_h} & \frac{\partial r_n(x)}{\partial R_w} \end{bmatrix} \quad (18)$$

$$J(x) = \begin{bmatrix} \frac{\partial r_1(x)}{\partial e} & \frac{\partial r_1(x)}{\partial q} & \frac{\partial r_1(x)}{\partial m} & \frac{\partial r_1(x)}{\partial h} & \frac{\partial r_1(x)}{\partial R_s} & \frac{\partial r_1(x)}{\partial R_h} & \frac{\partial r_1(x)}{\partial R_w} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \frac{\partial r_i(x)}{\partial e} & \frac{\partial r_i(x)}{\partial q} & \frac{\partial r_i(x)}{\partial m} & \frac{\partial r_i(x)}{\partial h} & \frac{\partial r_i(x)}{\partial R_s} & \frac{\partial r_i(x)}{\partial R_h} & \frac{\partial r_i(x)}{\partial R_w} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \frac{\partial r_n(x)}{\partial e} & \frac{\partial r_n(x)}{\partial q} & \frac{\partial r_n(x)}{\partial m} & \frac{\partial r_n(x)}{\partial h} & \frac{\partial r_n(x)}{\partial R_s} & \frac{\partial r_n(x)}{\partial R_h} & \frac{\partial r_n(x)}{\partial R_w} \end{bmatrix} \quad (19)$$

where: $r_i(x)$ – an i -th component of the residual vector. Some of the components in the $J(x)$ matrix are equal 0, e.g. if the given function doesn't depend on the given variable from the solution vector "x".

The constraints are defined in the form of the penalty functions. There are penalty function for the maximum contact stress for the external and internal sleeves. Additionally, there are two different penalty functions for the contact stress which arise in the external sleeve due to contact with cycloidal gear's lobes and pits. The contact stress in the external sleeves is calculated from equations (19) and (20) for the lobes and pits respectively. The contact stress in the internal sleeves is calculated from equation (21). Forms of contact are shown in Fig. 2.

$$\sigma_{EXLOBE} \sigma_{EXPOS} = 0.5642 \cdot \sqrt{\frac{\left(\frac{F_{EX}}{h}\right) \cdot (\rho + q)}{\eta \cdot \rho \cdot q}} \quad (20)$$

$$\sigma_{EXPIT} \sigma_{EXNEG} = 0.5642 \cdot \sqrt{\frac{\left(\frac{F_{EX}}{h}\right) \cdot (|\rho| - q)}{\eta \cdot |\rho| \cdot q}} \quad (21)$$

$$\sigma_{IN} = 0.5642 \cdot \sqrt{\frac{\left(\frac{F_{IN}}{h}\right) \cdot (R_h - R_s)}{\eta \cdot R_h \cdot R_s}}, \quad (22)$$

where: σ_{EXLOBE} – the contact stress in the external sleeve, which is in contact with the cycloidal wheel's lobe, σ_{EXPIT} – the contact stress in the external sleeve, which is in contact with the cycloidal wheel's pit, σ_{IN} – the contact stress in the internal sleeve, F_{EX} – maximum resulting force acting on the external sleeve given in equation (22), F_{IN} – the resulting force acting on the internal sleeve given in equation (23), ρ – radius of the cycloidal wheel's curvature at the point of contact with the external sleeves, η – elasticity coefficient given in equation (24).

$$F_{EX} = \frac{4M_h z_s}{e \cdot z_s' z_k} \quad (23)$$

$$F_{IN} = \frac{2M_h \cdot z_s' \cdot \cos \alpha}{\pi \cdot R_w} \quad (24)$$

$$\eta = \frac{1 - \nu_1^2}{E_1} + \frac{1 - \nu_2^2}{E_2} \quad (25)$$

where: ν_1, ν_2 – Poisson's ratios for the material of the cycloidal wheel and the sleeve respectively, E_1, E_2 – Young's modules for the material of cycloidal wheel and sleeve respectively, M_h – the input torque, α – the angle between an adjacent holes.

In the optimization process, the curvature radius is calculated for the extremal points of the cycloidal wheel. The Newton method presented in equation (25) is used for calculation of the angle corresponding to these locations. The equation for curvature radius was derived from the parametric equations of the cycloidal wheel and equation (26) using symbolical calculations in the MATLAB software and numerical differentiation.

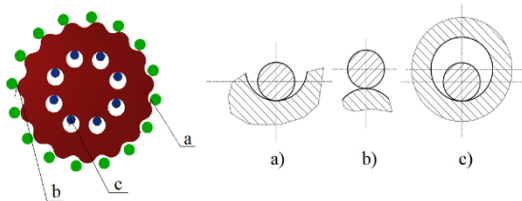


Fig. 2. Forms of the contact between the cycloidal wheel and the sleeves. The external sleeve-pit contact (a), the external sleeve-lobe contact (b) and the cycloidal wheel hole-internal sleeve contact (c)

$$\alpha_{NEXT} = \alpha_{PREV} - \frac{\frac{d\rho(\alpha)}{d\alpha}}{\frac{d}{d\alpha}\left(\frac{d\rho(\alpha)}{d\alpha}\right)}, \quad (26)$$

$$\rho(\alpha) = \frac{(u'(\alpha)^2 + v'(\alpha)^2)^{\frac{3}{2}}}{|v(\alpha)''u(\alpha)' - u(\alpha)''v(\alpha)'|} \quad (27)$$

where: $u(\alpha), v(\alpha)$ – parametric equations of the cycloidal wheel, α – the angle as a parameter in equations $u(\alpha)$ and $v(\alpha)$. The constraints are also defined for the limitations of the curvature radius. The components of $J(x)$ matrix for the curvature radius penalty functions are given in equation (27).

$$J_{\rho_{min,e}}(x) = 2 \cdot (\rho_{min} - \rho) \cdot \frac{\partial \rho(\alpha_{max})}{\partial e} \quad (28)$$

where: $J_{\rho_{min,e}}$ – the component of the Jacobi matrix which correspond to the penalty function in the residual vector, which establishes lower limit of the positive curvature radius at the lobe. This is first component in a row, which corresponds to the eccentricity variable derivative. Other components contain derivatives by "q" (radius of the external sleeve) and "m" (short-width coefficient) and the others in a row are equal 0. α_{max} – the angle for the extremal curvature radius value calculated by the Newton method.

5 The optimization results

The gradient-type optimization methods are intended to find the local extremes of the objective function. The Steepest Descent Method is slower, but gives more precise

results which are dependent of the scale coefficients. The Levenberg-Marquardt algorithm is much faster, but it can stop due to unfavorable conditions of the objective function. These algorithms were tested for the various starting points. Results of the optimization are presented in tables 1 and 2 and in the plots of the gear contours in Figure 3.

Table 1. Optimization results (the stress and the volume). N – number of optimization variant, BOPT – before the optimization, LM – Levenberg Marquardt Algorithm, SD – Steepest Descent Method, VOL – the volume of the cycloidal gearbox, ITER – number of iterations of the optimization algorithm, σ_{EXPOS} – stress at the lobe, σ_{EXNEG} – stress at the pit, σ_{IN} – stress at the internal sleeve

N	Method	Figure	σ_{EXPOS}	σ_{EXNEG}	σ_{IN}	VOL	ITER
1	BOPT		424	139	303	1.1718e+05	
1	LM	3(1)	488	177	509	6.0680e+04	127
1	SD	3(2)	476	160	341	8.6033e+04	8249
2	BOPT		855	331	552	3.0328e+04	
2	LM		685	466	491	8.8429e+04	22
2	SD		717	486	547	8.4566e+04	96
3	BOPT		419	142	303	1.2661e+05	
3	LM		490	175	505	6.0779e+04	109
3	SD		478	164	347	8.8072e+04	8835
4	BOPT		415	146	303	1.3643e+05	
4	LM		489	185	505	6.5570e+04	31
4	SD		480	169	352	9.0274e+04	9265

Table 2. Optimization results (the solution vector): N – number of optimization variant, M - optimization method, SP – starting point, LM – Levenberg Marquardt Algorithm, SD – Steepest Descent Method

N	M	Fig.	e	q	m	h	Rs	Rh	Rw
1	SP		2.8	3.5	0.7	10	5	8	30
1	LM	3(1)	2.3317	4.1502	0.6238	8.2701	3.5290	12.4852	34.784
1	SD	3(2)	2.4637	3.5488	0.6747	9.1856	4.8105	8.3151	30
2	SP		1.5	2	0.65	8	3	5	20
2	LM		1.6515	2.3424	0.4513	8.8444	3.0593	4.9149	21.178
2	SD		1.7828	2.1442	0.4764	8.0453	3.0142	4.9958	20.001
3	SP		2.9	3.4	0.7	10	5	8	30
3	LM		2.2757	4.1670	0.6240	8.4674	3.4925	11.8874	34.31
3	SD		2.5077	3.4718	0.6741	9.0349	4.7901	8.3506	30
4	SP		3	3.3	0.7	10	5	8	30
4	LM		2.3225	4.0285	0.6166	8.3739	3.5707	11.9415	33.681
4	SD		2.5542	3.3977	0.6735	8.8798	4.7723	8.3817	30

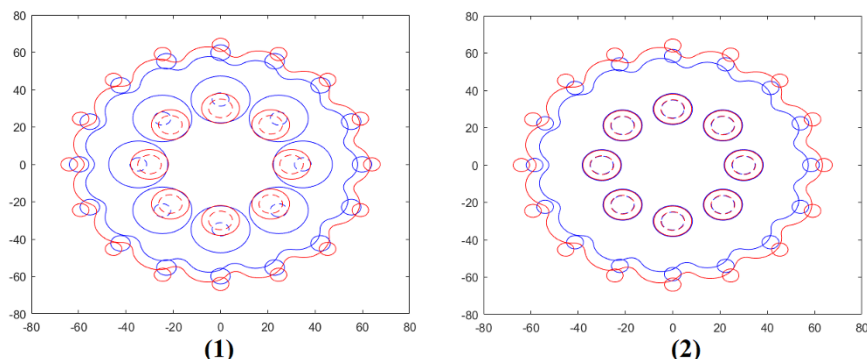


Fig. 3. The cycloidal gearbox shapes (1-2). Red line – before the optimization, blue line – after the optimization, dashed line – internal sleeves. Starting points are given in tables 1 and 2

6 Conclusions

Presented gradient optimization methods lead to smaller gearbox volume decrease compared with methods based on the genetic algorithms [7]. Gradient based methods stop in the local minimum of the objective function. If the optimized function has many local minima on the given area, then it is difficult to obtain large decrease in its value. Optimization should be started from the various starting points.

In the presented results, volume of the cycloidal gearbox decreased in the optimization variants 1, 3, 4 and increased in the optimization variant 2. In the variant 2, the starting point was placed at the domain of the penalty function, where the contact stress limit (400 MPa) was significantly exceeded. The contact stress at the lobes of the cycloidal wheel for the starting point in variant 2 is 855 MPa. The Levenberg-Marquardt algorithm stopped after 22 iterations finding minimum which is placed in the domain of the penalty function ($\sigma_{\text{EXLOBE}}=685$ MPa).

In the optimization variants 5 and 6 starting point is placed near to the boundary of the penalty function domain and the results obtained in the Steepest Descent Method and in the Levenberg-Marquardt algorithm are similar.

For the step coefficient $\alpha=10^{-13}$ in the Steepest Descent Method, number of steps is 80-times greater than in the Levenberg-Marquardt algorithm. In most variants of optimization, both methods give similar results in which contact stress at the lobes differ by about 20 MPa. The Steepest Descent Method doesn't change R_w parameter and the other parameters concerning the internal sleeves and the holes (R_s and R_h) change by 4.6% and 4.8% respectively in the optimization process. Using the Levenberg-Marquardt algorithm the same values change by 15.9% (R_w), 30.15% (R_s) and 56% (R_h).

First attempts of the optimization without curvature constraints lead to the shapes of the cycloidal wheel which are useless in the technical applications. It is very important to constrain negative and positive curvature radiuses. It is also important to apply box constraints on the parameters of the solution vector to obtain shapes of the gear which work properly.

The optimization variants 1, 3 and 4 are characterized by significant reduction of the cycloidal gearbox mass. Both methods can be used to improve the manufacturing process, but the Levenberg-Marquardt algorithm is faster and lead to better designs of the cycloidal gearbox.

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